

WWW 2026 COMPANION — ONLINE SESSION

Quantum Computing, Data Standard, and Blockchain Interoperability for the Web

Standards, Governance, and Sustainability for the Next-Generation Web

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Quantum-Safe, Efficient, and AI-Enhanced Blockchains for the Web: A Cooperative Tutorial on Quantum Computing, Blockchain Applications, and Data Standards



Quantum-Safe



Data Standards



Interoperability



Sustainability

Agenda

01

Quantum-Safe Blockchain

NIST post-quantum cryptography standards (FIPS 203–206) and global migration roadmaps including China, US, EU, and India

~6 min

02

Open Data Standards

Croissant metadata standard, NeurIPS 2026 requirements, and blockchain data as open science

~5 min

03

Interoperability & AI Ethics

IEEE 7000-series standards review and global AI ethics policy developments across China, Europe, and US

~6 min

04

Closing Synthesis

Three pillars, one foundation: sustainability by design as a principle woven through all three

~5 min

The Bridge

The preceding talks introduced powerful individual technologies. This talk bridges them into a **unified foundation** for the next-generation Web.

Quantum-Safe Infrastructure

Cryptographic foundations that endure the quantum threat

NIST FIPS 203-206 standards, global migration roadmaps, Algorand PQC roadmap targeting 2027

Open Data Standards

Information layer that is accessible and reproducible

Croissant metadata standard, Beacon Chain and Uniswap indices in Nature Scientific Data, NeurIPS 2026 RAI requirements

Accountable Interoperability

Governance layer that is transparent and ethical

ERC-8004 vs. Google A2A governance analysis, IEEE 7000-series standards review across China, EU, and US

✔ Sustainability by Design

Not a separate pillar, but a principle woven through all three — embedding environmental accountability into standards and protocols from the start

Liu, Dongping, Aoyu Zhang, and Luyao Zhang. "AI-Enhanced Quantum-Safe Blockchains for the Web: A Cooperative Tutorial on Quantum Computing, Blockchain Applications, and Data Standards." Companion Proceedings of the ACM Web Conference 2026 (WWW Companion '26). ACM, 2026.

<https://dl.acm.org/doi/10.1145/3774905.3793916>

The Quantum Threat to Blockchain

Breaking RSA and ECC

A cryptographically relevant quantum computer (CRQC) can break **RSA and ECC** — the cryptographic foundations of blockchain signatures and consensus mechanisms.

Harvest Now, Decrypt Later

Adversaries collect encrypted blockchain data **today** to decrypt when quantum arrives. Long-lived digital assets face existential risk **before** a CRQC even exists.

Implications for Blockchain

Bitcoin UTXOs, smart contract state, and all digitally signed assets rely on ECC/RSA. The question is not *if* but *when* — and standards bodies have already responded.

The Standards Response (2016–2024)

- **2016** — NIST initiates Post-Quantum Cryptography standardization competition
- **2022** — First algorithm selections announced (CRYSTALS-KYBER, CRYSTALS-DILITHIUM, FALCON, SPHINCS+)
- **2024** — FIPS 203, 204, 205 finalized (August 13). FIPS 206 (FN-DSA) expected 2025.

NIST Post-Quantum Cryptography Standards (August 2024)

Standard	Algorithm	Purpose	Key Characteristics
FIPS 203 ML-KEM	Module-Lattice-Based Key-Encapsulation Mechanism	Replaces RSA/ECDH for key exchange and key establishment	Based on CRYSTALS-KYBER; MLWE hardness assumption; quantum-resistant key encapsulation
FIPS 204 ML-DSA	Module-Lattice-Based Digital Signature Algorithm	Primary replacement for ECDSA/RSA digital signatures	Based on CRYSTALS-DILITHIUM; MLWE + MSIS; primary signature standard for most use cases
FIPS 205 SLH-DSA	Stateless Hash-Based Digital Signature Algorithm	Hash-based backup for algorithmic diversity and trust	Based on SPHINCS+; minimal security assumptions; larger signatures but conservative design
FIPS 206 FN-DSA	FALCON-based Digital Signature Algorithm	Smaller signatures for constrained environments	Based on FALCON; NTRU lattice assumptions; compact signatures; expected 2025

Security foundation: All four standards are based on the **Module Learning With Errors (MLWE)** problem — believed to be hard for both classical and quantum computers. **Trade-off:** larger key/signature sizes (e.g., ML-DSA-65 signatures: 3,293 bytes vs. 64 bytes for ECDSA) but provably quantum-resistant.

References

- [1] NIST. "FIPS 203: Module-Lattice-Based Key-Encapsulation Mechanism Standard." August 13, 2024. <https://csrc.nist.gov/pubs/fips/203/final>
- [2] NIST. "FIPS 204: Module-Lattice-Based Digital Signature Standard." August 13, 2024. <https://csrc.nist.gov/pubs/fips/204/final>
- [3] NIST. "FIPS 205: Stateless Hash-Based Digital Signature Standard." August 13, 2024. <https://csrc.nist.gov/pubs/fips/205/final>
- [4] NIST. "FIPS 206: FN-DSA" (expected 2025). <https://csrc.nist.gov/projects/post-quantum-cryptography>

Global Quantum-Safe Migration Roadmaps

Jurisdiction	Timeline	Key Milestones	Distinctive Approach
United States (CNSA 2.0)	2025–2035	Software/firmware PQC signatures by 2025; legacy systems deprecated by 2030; full migration by 2035	Military-first approach; NSA-led mandate for national security systems
European Union (ENISA)	2026–2035	National roadmaps by Dec 2026; critical infrastructure protected by 2030; full transition by 2035	Coordinated member-state approach; ENISA harmonized guidelines
United Kingdom (NCSC)	2028–2035	Discovery by 2028; high-priority migration by 2031; full adoption by 2035	Risk-prioritized phased migration; NCSC practical guidance
India (DST/NQM)	2027–2033	CII foundations by 2027; high-priority by 2028; full PQC by 2029 (CII) / 2033 (enterprises)	Accelerated timeline for critical infrastructure; national quantum mission aligned
Australia (ASD)	2026–2030	Traditional asymmetric cryptography deprecated by end of 2030	Most aggressive timeline; ASD direct mandate for government systems
China (SCA/TC260)	2026–2035	"Three-five-ten" roadmap; PQC GB/T by 2026; dual-stack 2027–2028; 80%+ by 2030	"PQC baseline + QKD enhancement" dual-track; GB/T 42829-2023, GB/T 43692-2024; structural gap noted

China's distinctive approach: A "PQC baseline + QKD enhancement" dual-track strategy. Strong QKD engineering standards (GB/T 42829-2023, GB/T 43692-2024) but PQC national standards still under development — a structural gap of "engineering leading while standards lag."

References

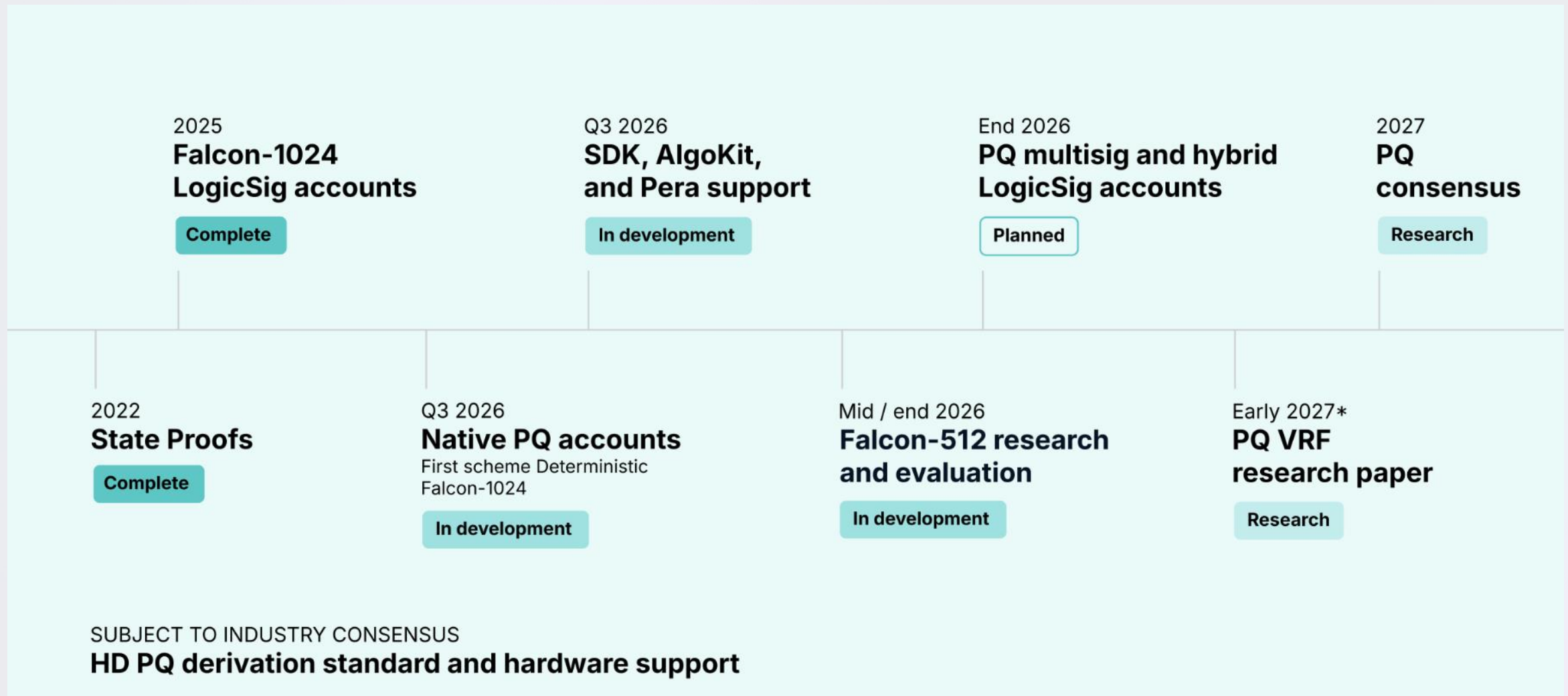
[1] NSA. "Commercial National Security Algorithm Suite 2.0 (CNSA 2.0)." September 2022. <https://www.nsa.gov/Press-Room/Fact-Sheets/Article/3949550/cnsa-20/>

[2] European Commission. "Coordinated Implementation Roadmap for Post-Quantum Cryptography." 2025. <https://digital-strategy.ec.europa.eu/en/coordinated-implementation-roadmap-post-quantum-cryptography>

[3] UK NCSC. "Post-Quantum Cryptography Roadmap." 2025. <https://www.ncsc.gov.uk/collection/post-quantum-cryptography-roadmap>

[4] GSMA. "Post Quantum Government Initiatives by Country and Region." February 2026. <https://www.gsma.com/newsroom/post-quantum-government-initiatives-by-country-and-region/>

Algorand's Post-Quantum Roadmap (by 2027)



[1] Algorand Foundation. "Algorand Post-Quantum Cryptography Roadmap." June 18, 2026. <https://algorand.co/blog/algorand-post-quantum-cryptography-roadmap>

Algorand's Post-Quantum Roadmap (by 2027)

Algorand Foundation published its PQC roadmap on June 18, 2026. Recognized by Google Quantum AI as among the limited set of smart-contract platforms with real-world PQC deployment.

Q3 2026

Native Falcon-1024 accounts — ledger-native, not LogicSig-based

Cryptographic agility — network-level support for multiple concurrent signature schemes
Enables integration of Ed25519, Falcon, ML-DSA with minimal protocol disruption

End of 2026

Falcon-512 support — smaller key sizes: 897-byte public key vs. 1,793 bytes for Falcon-1024

Post-quantum multisig for multi-cryptography schemes
ECC + lattice hybrid signatures

Ongoing Research

Post-quantum VRF (Verifiable Random Function) for committee selection

Cryptographic agility framework enabling future NIST standards
Target: broad quantum resilience by 2027

 **Key Insight:** Cryptographic agility — the ability to support multiple signature schemes concurrently — is the foundation for quantum-safe blockchain migration, not any single algorithm.

[1] Algorand Foundation. "Algorand Post-Quantum Cryptography Roadmap." June 18, 2026. <https://algorand.co/blog/algorand-post-quantum-cryptography-roadmap>

[2] Google Quantum AI. "Safeguarding Cryptocurrency by Disclosing Quantum Vulnerabilities Responsibly." 2026.

Open Data Standards: Three Ecosystems, One Principle

Open data is necessary but not sufficient — **structured metadata + open access + reproducibility** make it scientific infrastructure. Common principle: **FAIR** (Findable, Accessible, Interoperable, Reusable).

Geospatial (STAC)

SpatioTemporal Asset Catalog standard for Earth observation data

Callable via MCP (Model Context Protocol) as demonstrated by Dongping Liu — open geospatial data becomes agent-callable scientific infrastructure

ML Datasets (Croissant)

JSON-LD metadata standard for ML-ready datasets from Google Research

NeurIPS 2026 now mandates Croissant as a submission requirement with Responsible AI metadata fields

Blockchain Data

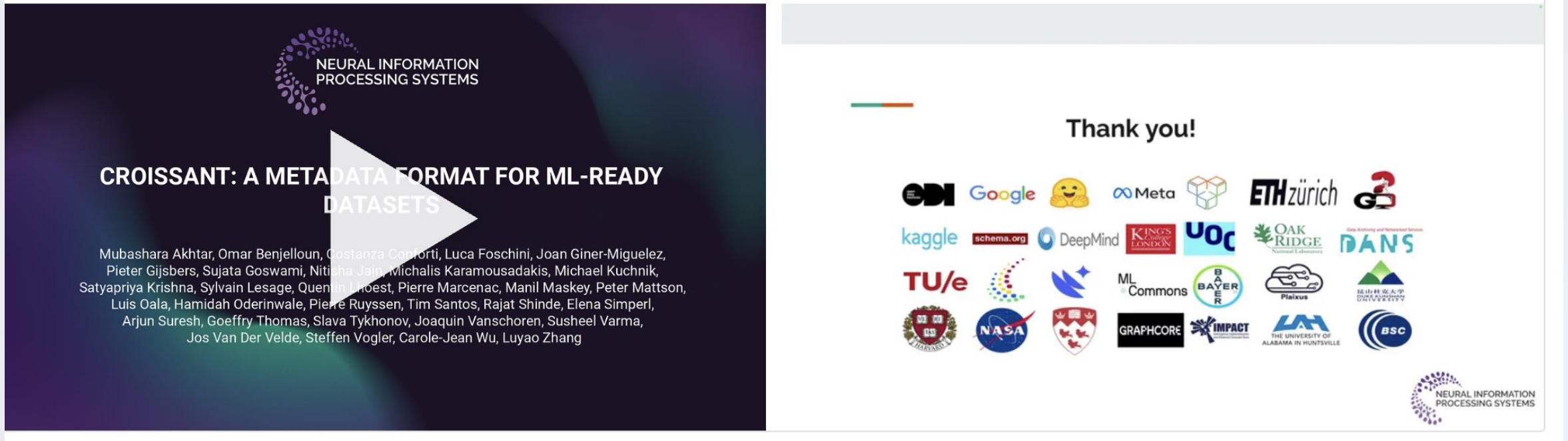
Ethereum Beacon Chain and Uniswap Daily Transaction Indices

Both published in Nature's Scientific Data (2025) — blockchain data recognized as first-class scientific data when properly standardized

Findable	Accessible
Interoperable	Reusable

Croissant and the NeurIPS 2026 Dataset Requirements

Croissant (Akhtar et al., NeurIPS 2024) provides a uniform JSON-LD metadata schema for ML datasets. The NeurIPS 2026 Evaluations & Datasets Track now mandates three requirements:



- [1] Akhtar, Omar et al. "Croissant: A Metadata Standard for ML-Ready Datasets." NeurIPS, 2024. <https://proceedings.neurips.cc>
- [2] "NeurIPS Evaluations & Datasets 2026 Reviewing Guidelines — Requirements for Datasets." <https://neurips.cc/Conferences/2026/EvaluationsDatasetsReviewerGuidelines>

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Submissions with a Dataset contribution

If a submission introduces a new dataset, regardless of the contribution type, please evaluate it using both the provided automated tools and your expert judgment.

Meta-data requirements: *All submissions that contribute datasets are required to provide a valid Croissant file.* Following the full paper submission deadline, we conducted compliance checks and desk-rejected submissions that clearly failed to meet this requirement. However, some non-compliant cases may have remained undetected. If you encounter a submission missing a Croissant file or providing an invalid/placeholder one, please flag it to your Area Chair.

Dataset Accessibility: *Datasets should be available to all reviewers, ACs, and SACs at the time of submission, and without a personal request to the PI.* We check whether we can access the dataset automatically based on the Croissant meta-data, but it is possible that data is still manually accessible to you. In rare cases, a dataset may require credentialized (gated) access when this is necessary for ethical or public-interest reasons and is implemented through an established and efficient credentialization process. Authors may provide access instructions with the submission or in response to a request via OpenReview. Please do not engage in substantive discussion with the authors beyond requesting access details. If access cannot be obtained, or if the use of credentialized access appears inconsistent with the policy, please notify your Area Chair.

Meta-data completeness: *Authors were asked to include a minimal set of Croissant and Responsible AI (RAI) metadata as described in the E&D data hosting guidelines.* To facilitate the review process, you will be provided with an automated dataset reviewer report generated from the submission's Croissant file. This will be posted on the paper's OpenReview page. This report checks technical compliance, such as URL accessibility, hosting platform validity, and the presence of core and RAI metadata.

Significance: Croissant + RAI metadata transforms dataset submission from a technical upload into an **accountable, transparent scientific artifact** — the same principle should apply to quantum and blockchain datasets.

ⓘ [1] Akhtar, Omar et al. "Croissant: A Metadata Standard for ML-Ready Datasets." NeurIPS, 2024. <https://proceedings.neurips.cc>

[2] "NeurIPS Evaluations & Datasets 2026 Reviewing Guidelines — Requirements for Datasets."

<https://neurips.cc/Conferences/2026/EvaluationsDatasetsReviewerGuidelines>

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1	2	3
Croissant Metadata All dataset submissions must provide a valid Croissant file (JSON-LD format) Automated compliance checking via reviewer report on OpenReview Checks: URL accessibility, hosting platform validity, core metadata presence	Open Access Hosting Datasets must be accessible to all reviewers without requiring a personal request to the PI Open access is the default Credentialized access permitted only for ethical/public-interest reasons through established processes	Responsible AI (RAI) Fields Authors must include RAI metadata covering: • Data limitations and known biases • Personal/sensitive information • Intended use cases • Social impact assessment

Significance: Croissant + RAI metadata transforms dataset submission from a technical upload into an **accountable, transparent scientific artifact** — the same principle should apply to quantum and blockchain datasets.

- ❖ [1] Akhtar, Omar et al. "Croissant: A Metadata Standard for ML-Ready Datasets." NeurIPS, 2024. <https://proceedings.neurips.cc>
- [2] "NeurIPS Evaluations & Datasets 2026 Reviewing Guidelines — Requirements for Datasets." <https://neurips.cc/Conferences/2026/EvaluationsDatasetsReviewerGuidelines>

Blockchain Data as Open Science (Scientific Data 2025)

NATURE SCI. DATA

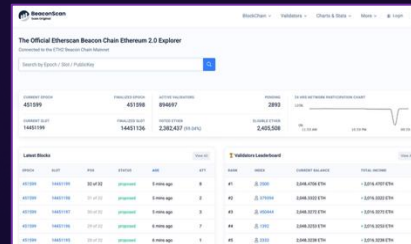
Ethereum Beacon Chain Data Engineering Framework

Yan, Qin, Wanlin Deng, Yutong Quan, Xintong Wu, and **Luyao Zhang**

Structured extraction of Ethereum 2.0 consensus-layer data

Enables reproducible research on:

- Validator behavior and staking economics
- Slashings and attestations
- Consensus mechanism dynamics



NATURE SCI. DATA

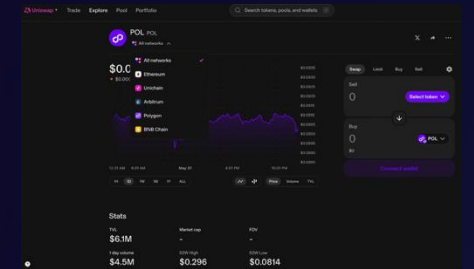
Uniswap Daily Transaction Indices

Chemaya, Valeriya, Nikita Kharlamov, Wanlin Deng, and **Luyao Zhang**

Daily aggregated indices from DEX transactions across multiple liquidity pools

Enables empirical DeFi research:

- Without full-node infrastructure
- With reproducible, standardized data
- Cross-pool comparative analysis

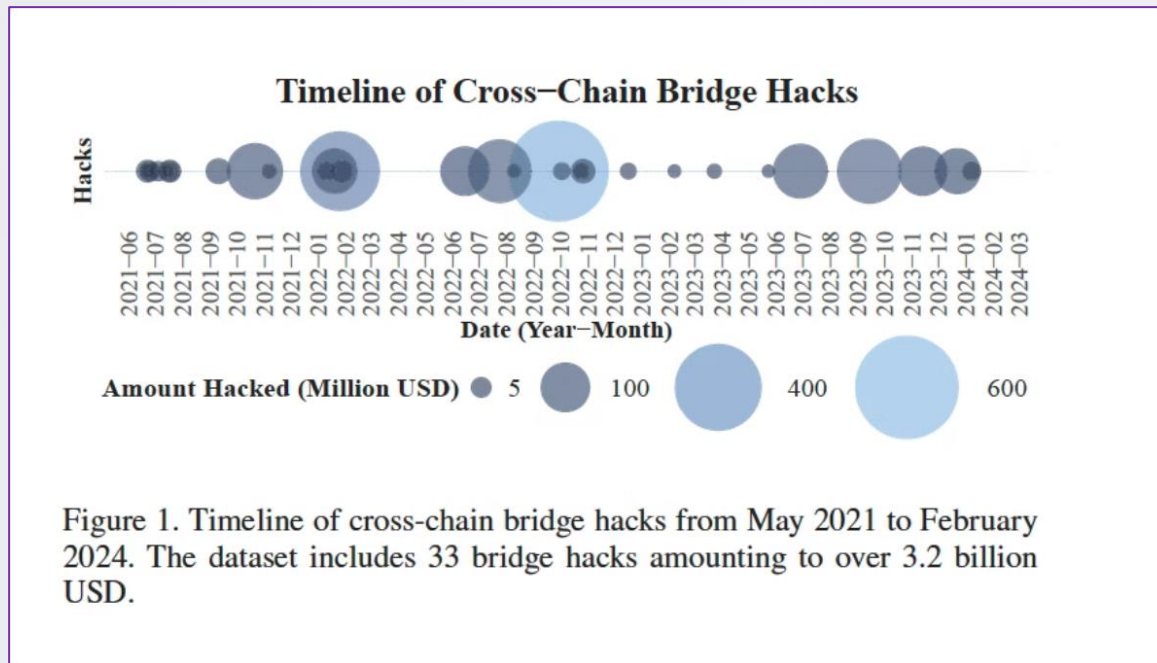


Key message: Both published in Nature's *Scientific Data* – blockchain data recognized as first-class scientific data when properly standardized. Both follow FAIR principles and would benefit from Croissant-style metadata + RAI fields.

[1] Yan, Qin, Wanlin Deng, Yutong Quan, Xintong Wu, and Luyao Zhang. "Ethereum Beacon Chain Data Engineering Framework." *Scientific Data* (Nature Portfolio), 2025.

[2] Chemaya, Valeriya, Nikita Kharlamov, Wanlin Deng, and Luyao Zhang. "Uniswap Daily Transaction Indices." *Scientific Data* (Nature Portfolio), 2025.

Blockchain Interoperability (IEEE S&P 2024)



Key Findings & Quantum Connection

Over \$2B lost to bridge hacks — interoperability introduces new attack surfaces

Custodial vs. non-custodial trade-offs mapped systematically

Quantum threat: As quantum computing threatens cryptographic foundations, cross-chain bridges must incorporate post-quantum cryptography

Systematic Review of Cross-Chain Bridges

Comprehensive taxonomy covering:

Hash Time-Locked Contracts (HTLC) — atomic swaps without trusted third parties

Notary schemes — trusted intermediaries validate cross-chain transactions

Relay chains — light client verification across chains

Sidechains — pegged two-way bridges with independent consensus

SoK contribution: First systematic taxonomy of blockchain interoperability security — bridges the gap between cryptographic theory and deployed system security.

Governance of AI Agent Interoperability (ICMO 2026)

Wang, Yutian and **Luyao Zhang**

ERC-8004

Permissionless, On-Chain, Community-Governed

"Discover, choose, and interact with agents across organizational boundaries without pre-existing trust"

Governance focus: Trust and security — "what to build and why"

Community: Higher discourse convergence on opinions despite decentralized participation

Google A2A

Corporate-Led, Open Protocol, Centrally Directed

"An open protocol enabling communication and interoperability between opaque agentic applications"

Governance focus: Documentation and delivery — "how to build and ship"

Community: Broader topic spread but comparable participation inequality

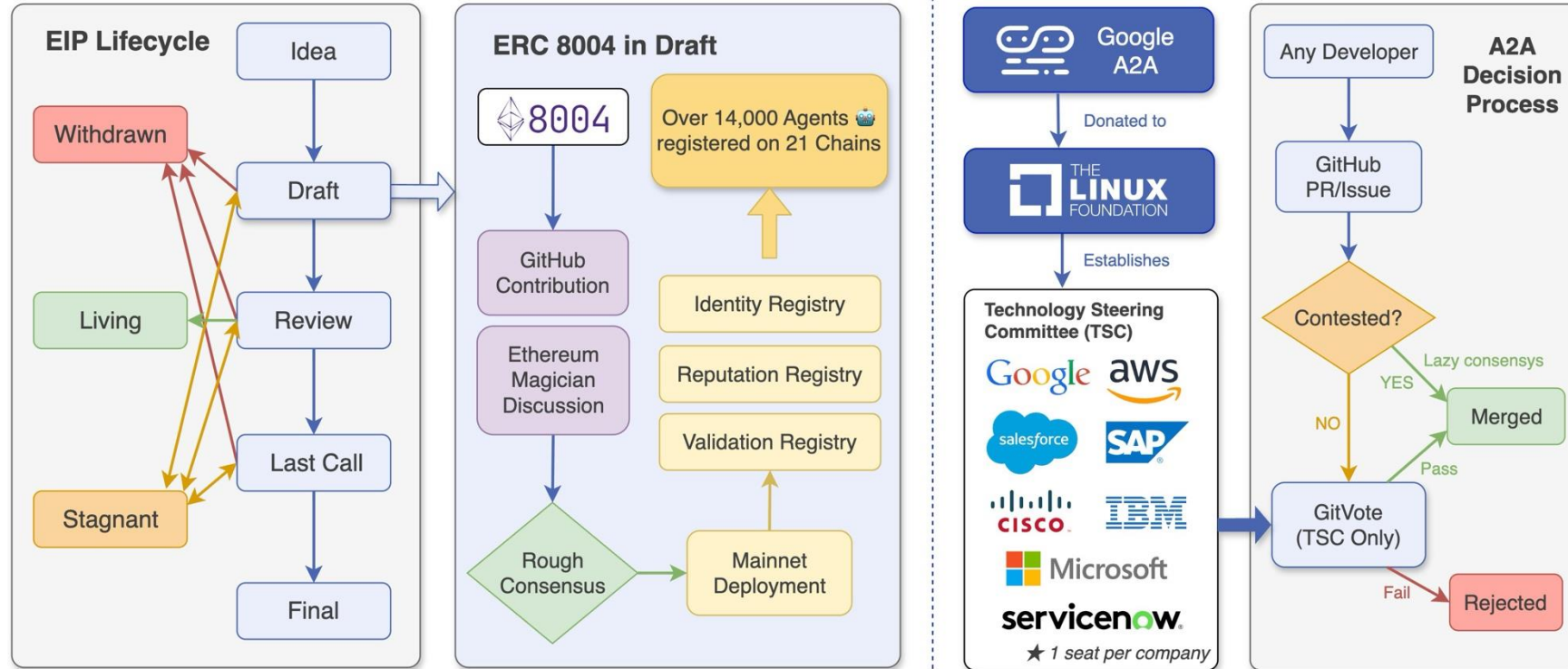
📄 **Method:** LLM-powered comparative pipeline — automated annotation + neural topic modeling + multi-layer network analysis on 4,323 governance participation records. **Key finding:** Discourse alignment is denser in permissionless settings; governance form influences thematic focus and power distribution.

[1] Wang, Yutian and Luyao Zhang. "Agentic Analysis for Agentic Infrastructure: An LLM-Powered Pipeline for Comparative Governance of DAO and Corporate AI Protocols." arXiv:2606.26203, 2026.

[2] Wu, Xintong, Wanlin Deng, Yutong Quan, Lin William Cong, and Luyao Zhang. "Trust Dynamics in Cryptocurrency Markets: Centralized vs. Decentralized Exchanges." arXiv:2404.17227, 2024 (rev. 2026).

ERC-8004 and Google A2A

Same problem (agent operability), different governance forms.



ERC-8004: “discover, choose, and interact with agents across organizational boundaries without pre-existing trust”

(<https://ethereum-magicians.org/t/erc-8004-trustless-agents/25098>)

Google A2A: “an open protocol enabling communication and interoperability between opaque agentic applications”

(<https://github.com/a2aproject/A2A>)



AI Ethics Standards: A Global Review (IEEE Shanghai, June 2026)

IEEE 7000-Series Standards

Standard	Scope	URL
IEEE 7000-2021	Model Process for Addressing Ethical Concerns During System Design	Link
IEEE 7001-2021	Transparency of Autonomous Systems	Link
IEEE 7002-2022	Data Privacy Process for AI Systems	Link
IEEE 7003-2024	Algorithmic Bias Considerations	Link
IEEE 7009-2024	Fail-Safe Design for Autonomous Systems	Link
IEEE 7010-2020	Wellbeing Metrics for Ethical AI	Link

2026 Global Policy Developments

China

Ten-department "AI Ethics Review and Service Measures" (April 2026). Six review dimensions (human wellbeing, fairness, transparency, privacy, accountability, controllability) mapped directly to IEEE 7000-series standards.

European Union

EU AI Act harmonized standards (hENs) delayed; no standard cited in Official Journal as of mid-2026. Creates a 12–18 month standards gap that IEEE voluntary standards can help bridge.

United States

Gartner predicts 15M+ enterprise AI agents by 2028 but only 13% with adequate governance. IEEE P7999 addresses this organizational ethics oversight gap.

 **The "Connective Layer" concept:** Standards provide the connective tissue between vertically focused technical requirements and horizontally oriented social outcomes. Technical content converges even where governance philosophies diverge — this is the foundation for cross-jurisdiction collaboration.

Note: Luyao Zhang serves as Vice-Chair, IEEE P7999 SG1 (Foundational Framework for Organizational AI Ethics Oversight)

[1-6] IEEE 7000-2021 through 7010-2020 series. <https://standards.ieee.org>

[7] China MIIT et al. "AI Ethics Review and Service Measures (Trial)." April 2026.

[8] EU AI Act (Regulation EU 2024/1689). <https://eur-lex.europa.eu/eli/reg/2024/1689>

Sustainability by Design: A Foundational Principle

Three resource-intensive technologies are converging on the next-generation Web — each carries significant environmental costs:

Quantum Computing

Cryogenic cooling of dilution refrigerators at temperatures near **15 millikelvin**

Significant energy footprint per qubit — must be accounted for in lifecycle design

Blockchain

Redundant validation across nodes, consensus mechanisms, cross-chain bridge operations

Energy-intensive proof-of-work/stake and multi-chain coordination

AI Agents

Large-scale LLM inference on **GPU clusters** at data center scale

Growing demand as agent deployments scale — Gartner predicts 15M+ by 2028

✔️ 🌱 Sustainability by Design

Embedding environmental accountability into **standards and protocols from the start**, not measuring damage after the fact.
Every pillar of the next Web must carry environmental accountability from day one.

E-Liability for Embodied AI — Standards Review & The Gap

Review of publicly listed embodied AI standards — every standard covers a piece of the puzzle, but **none address environmental liability as a supply-chain accounting problem**:

Standard	Scope	What It Covers
IEEE P3704	Evaluation Methods for Embodied Intelligence	Capability benchmarks
IEEE P3707	Embodied AI in Elderly Care	Deployment guidelines
IEEE P3927	Industrial Capability Requirements	Industrial assessment
IEEE P3927.1	Training Data Acquisition	Data management
IEEE P3960	System Architecture for Embodied Intelligence OS	OS-level design
IEEE P3584	Carbon Emission Measurement of AI Model Training	Training only — point-in-time
IEEE 7010	Wellbeing Impact Assessment	Human wellbeing metrics

✓ The Critical Gap

None of the existing standards address **E-Liability across the full mechatronic supply chain**:

Semiconductor fabrication → Robot assembly → Model development → Model training → Inference & deployment

E-Liability concept: Double-entry bookkeeping for environmental accounting — the same principle that prevents financial fraud can prevent environmental double-counting

📄 **Case study:** White paper proposal on "E-Liability of Embodied AI: Scaling Through the Mechatronic Supply Chain" (June 2026) — demonstrating how E-Liability transfers at each stage, from semiconductor fabrication through robot assembly to model training and inference.

Note: Luyao Zhang serves as Secretary, IEEE P3469 Working Group on Environmental Liability Process Models

All standards listed with public IEEE project page URLs. E-Liability concept: Kaplan, Robert S. and Karthik Ramanna. "Accounting for Climate Change." Harvard Business School Working Paper 21-104, 2021.

The Three Pillars: A Unified, Sustainable Foundation

Pillar	Technical Standard	Governance Standard	Sustainability by Design
Quantum-Safe Infrastructure <i>Security</i>	NIST FIPS 203–206 (ML-KEM, ML-DSA, SLH-DSA, FN-DSA)	Algorand PQC Roadmap (2027) Cryptographic agility, multi-scheme support	Cryogenic efficiency Hardware lifecycle accountability
Open Data Standards <i>Accessibility</i>	Croissant (NeurIPS 2024) Beacon Chain, Uniswap (Sci. Data 2025)	IEEE 7001 (Transparency) IEEE 7002 (Privacy) NeurIPS RAI metadata	Data carbon footprint Responsible AI fields
Interoperability & AI Ethics <i>Accountability</i>	ERC-8004, Google A2A SoK (IEEE S&P 2024)	IEEE P7999 (AI Ethics Oversight) IEEE 7003 (Bias) China AI Ethics Review	E-Liability (mechatronic chain) SolarChain marketplace

✔ **The Synergy**

Quantum-safe cryptography ensures the *security* of data and transactions. **Open data standards** ensure the *accessibility* and reproducibility of information. **Interoperability governance** ensures the *accountability* of protocols. **Sustainability by design** ensures the *endurance* of the infrastructure — measured, tracked, and verified from day one.

Ou, Shilin, Yifan Xu, Zhenshan Zhang, Luyao Zhang, and Ming-Chun Huang. "SolarChain: Bridging Physical Law, Verifiable Trust, and Sustainable Markets for Urban Energy Resilience." arXiv:2605.23162, 2026.

Key Takeaways & Call to Action

1 Quantum-safe infrastructure is being built now.

NIST standards are finalized (August 2024). Algorand targets quantum resilience by 2027. China's three-five-ten roadmap is underway. The migration is standards-driven, global, and urgent.

2 Open data standards make science reproducible and accountable.

Croissant is now a NeurIPS submission requirement with RAI metadata fields. Blockchain data (Beacon Chain, Uniswap) is first-class scientific data when properly standardized.

3 Interoperability is governance, not just protocol.

ERC-8004 vs. Google A2A shows that protocol design shapes power distribution. The IEEE 7000-series and China's ten-department framework converge on the same ethical dimensions.

4 Sustainability must be by design, not afterthought.

From cryogenic cooling to GPU inference to cross-chain validation — every layer of the next Web must embed environmental accountability.

5 The three pillars together

— quantum-safe infrastructure, open data standards, and accountable interoperability — collaboratively build the foundation for a Web that is secure, accessible, and sustainable.

Calls to Action

- Follow NIST FIPS 203–206 for your quantum-safe migration planning
- Explore IEEE 7000-series standards for ethically aligned AI design
- Adopt Croissant + RAI metadata for your datasets
- Learn more: github.com/QuantBlockchain/www26-qab4web

Liu, Dongping, Aoyu Zhang, and Luyao Zhang. "AI-Enhanced Quantum-Safe Blockchains for the Web: A Cooperative Tutorial on Quantum Computing, Blockchain Applications, and Data Standards." Companion Proceedings of the ACM Web Conference 2026 (WWW Companion '26). ACM, 2026. <https://dl.acm.org/doi/10.1145/3774905.3793916>

Thank You — Questions?

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